S25 Stat 134 Practice 2 Midterm *Solution*

Print your name: _____

SID Number: _____

Exam Information and Instructions:

- You will have 110 minutes to take this exam.
- The exam will be scored out of 50 points.
- Closed book/notes, calculator, or collaborating.
- To receive full credit, you must show your work or brief reasoning in every answer.
- Unless stated otherwise, **you may leave answers as unsimplified numerical and algebraic expressions**, and in terms of the Normal c.d.f. Φ. Finite sums are fine, but simplify any infinite sums.
- We will be using Gradescope to grade this exam. Write any work you want graded on the front of each page, in the space below each question.
- Do **NOT** write your final solution on a scratch paper. It won't be graded. If additional space is required, please utilize the blank page located at the end of this exam paper. When doing so, please write the phrase "Solution continues at page X" at the bottom of the page. Additionally, please indicate on the blank page which question you are continuing your solution for.

I certify that all materials in the enclosed exam are my own original work.

Sign your name: _____

1. [10 pts]**(Markov, complement rule)**

   (a) Suppose that there is a machine that gives out a random number Y between 0 and 80. You are also told that $\mathbb{E}[Y] = 20$. Now someone proposes you a game where you win if the number that shows up is strictly smaller than 40. Assume that you always play games when you have a chance of at least $\frac{1}{2}$ of winning. Given the information you have, can you determine whether you should agree to play the game?

   (b) Suppose that there is a machine that gives out a random number X between 20 and 100. You are also told that $\mathbb{E}[X] = 40$. Now someone proposes you a game where you win if the number that shows up is strictly smaller than 60. Assume that you always play games when you have a chance of at least $\frac{1}{2}$ of winning. Given the information you have, can you determine whether you should agree to play the game?

   Solution

   (a)

   $$\mathbb{P}(Y < 40) \geq \frac{1}{2}$$

   hence you should agree to play the game.

   (b)

   $$\mathbb{P}(X < 60) \geq \frac{1}{2}$$

   hence you should agree to play the game

2. [10 pts]**(hypergeometric, expectation and variance with indicators)** Consider a grid of n^2 cups, with n rows of n cups. Toss n ping pong balls at random into these cups, where at most one ball can occupy a particular cup. Let X be the number of unoccupied rows (i.e., where every cup in that row contains no balls). Find:

   (a) $P(X \geq 1)$;

   (b) $E(X)$;

   (c) $Var(X)$.

Solution

(a)

$$P(X \geq 1) = 1 - \frac{\binom{n}{1}^n}{\binom{n^2}{n}}$$

(b)

$$E(X) = n\frac{\binom{n}{0}\binom{n^2-n}{n}}{\binom{n^2}{n}}$$

(c)

$$Var(X) = n\frac{\binom{n}{0}\binom{n^2-n}{n}}{\binom{n^2}{n}} + n(n-1)\cdot\frac{\binom{n^2-2n}{n}}{\binom{n^2}{n}} - \left(n\frac{\binom{n}{0}\binom{n^2-n}{n}}{\binom{n^2}{n}}\right)^2$$

3. [10 pts]**(poisson and normal approx)** Consider a box containing 1000 balls, of which m are gold and the remaining $1000 - m$ are blue (Go Bears!). You draw 1000 balls from the box, randomly with replacement. Let X be the number of gold balls you get.

   (a) For any $1 \leq m \leq 1000$, what is $P(X = m)$? *Your answer should be in terms of m.*

   (b) For $m = 500$, what is $P(X = 500)$, approximately?

   (c) For $m = 2$, what is $P(X = 2)$, approximately?

Solution

(a)
$$P(X = m) = \binom{1000}{m} p^m (1-p)^{1000-m}.$$

(b)
$$P(X = 500) \approx \Phi\left(\frac{500 + 0.5 - 500}{\sqrt{250}}\right) - \Phi\left(\frac{500 - 0.5 - 500}{\sqrt{250}}\right).$$

(c)
$$P(X = 2) \approx e^{-2}\frac{2^2}{2!} = 2e^{-2}.$$

4. [10 pts]**(geometric distribution application)** Suppose that Player A and Player B take turns rolling a pair of balanced dice and that the winner is the first player who obtains the sum of 7 on a given roll of the two dice. Assume that the rolls are independent. What is the probability that Player A wins the game if he rolls first?

Solution

$$\mathbb{P}(W) = \frac{1}{6 - \frac{25}{6}}.$$

5. [10 pts]**(poisson, gamma, memoryless property)** NASA counts the arrivals of comets in 2023, which arrive according to a Poisson process with average rate $\lambda = 5/12$ per month.

   (a) (5 pts) Find the probability that the third comet in 2023 arrives sometime in the summer months (June, July, August, the 6-8th months of the year). [Warning: Be careful with counting from the beginning vs the end of the months!]

   (b) (5 pts) Suppose we see one comet in January and two comets in February. Find the expected total time of the arrival of the fourth comet of the year.

Solution

(a)

$$\mathbb{P}(5 \le T_3 \le 8) = \sum_{k=0}^{2} e^{-5\cdot 5/12} \frac{(5 \cdot 5/12)^k}{k!} - e^{-8\cdot 5/12} \frac{(8 \cdot 5/12)^k}{k!}.$$

(b)

$$\frac{22}{5}.$$

Blank page. Please indicate which question you are answering for.

Discrete

name and range	$P(k) = P(X = k)$ for $k \in$ range	mean	variance
uniform on $\{a, a+1, ..., b\}$	$\frac{1}{b-a+1}$	$\frac{a+b}{2}$	$\frac{(b-a+1)^2-1}{12}$
Bernoulli(p) on $\{0, 1\}$	$P(1) = p$; $P(0) = 1-p$	p	$p(1-p)$
binomial(n,p) on $\{0, 1, ..., n\}$	$\binom{n}{k} p^k (1-p)^{n-k}$	np	$np(1-p)$
Poisson(μ) on $\{0, 1, 2, ...\}$	$\frac{e^{-\mu}\mu^k}{k!}$	μ	μ
hypergeometric(n,N,G) on $\{0, ..., n\}$	$\frac{\binom{G}{k}\binom{N-G}{n-k}}{\binom{N}{n}}$	$\frac{nG}{N}$	$n\left(\frac{G}{N}\right)\left(\frac{N-G}{N}\right)\left(\frac{N-n}{N-1}\right)$
geometric(p) on $\{1, 2, 3, ...\}$	$(1-p)^{k-1}p$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
geometric(p) on $\{0, 1, 2, ...\}$	$(1-p)^k p$	$\frac{1-p}{p}$	$\frac{1-p}{p^2}$
negative binomial(r,p) on $\{0, 1, 2, ...\}$	$\binom{k+r-1}{r-1} p^r (1-p)^k$	$\frac{r(1-p)}{p}$	$\frac{r(1-p)}{p^2}$

Continuous

† undefined

name	range	density $f(x)$ for $x \in$ range	c.d.f $F(x)$ for $x \in$ range	Mean	Variance
uniform(a,b)	(a, b)	$\frac{1}{b-a}$	$\frac{x-a}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
normal(0,1)	$(-\infty, \infty)$	$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$	$\Phi(x)$	0	1
normal(μ,σ^2)	$(-\infty, \infty)$	$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}(x-\mu)^2/\sigma^2}$	$\Phi(\frac{x-\mu}{\sigma})$	μ	σ^2
exponential(λ) = gamma(1,λ)	$(0, \infty)$	$\lambda e^{-\lambda x}$	$1 - e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
gamma(r,λ)	$(0, \infty)$	$\Gamma(r)^{-1} \lambda^r x^{r-1} e^{-\lambda x}$	$1 - e^{-\lambda x} \sum_{k=0}^{r-1} \frac{(\lambda x)^k}{k!}$ for integer r	$\frac{r}{\lambda}$	$\frac{r}{\lambda^2}$
chi-square(n) = gamma($\frac{n}{2}$,$\frac{1}{2}$)	$(0, \infty)$	$\Gamma(\frac{n}{2})^{-1} (\frac{1}{2})^{\frac{n}{2}} x^{\frac{n}{2}-1} e^{-\frac{x}{2}}$	as above for $\lambda = \frac{1}{2}$, $r = \frac{n}{2}$ if n is even	n	$2n$
Rayleigh	$(0, \infty)$	$x e^{-\frac{1}{2}x^2}$	$1 - e^{-\frac{1}{2}x^2}$	$\sqrt{\frac{\pi}{2}}$	$\frac{4-\pi}{2}$
beta(r,s)	$(0, 1)$	$\frac{\Gamma(r+s)}{\Gamma(r)\Gamma(s)} x^{r-1}(1-x)^{s-1}$	see Exercise 4.6.5 for integer r and s	$\frac{r}{r+s}$	$\frac{rs}{(r+s)^2(r+s+1)}$
Cauchy	$(-\infty, \infty)$	$\frac{1}{\pi(1+x^2)}$	$\frac{1}{2} + \frac{1}{\pi} \arctan(x)$	†	†
arcsin = beta(1/2,1/2)	$(0, 1)$	$\frac{1}{\pi\sqrt{x(1-x)}}$	$\frac{2}{\pi} \arcsin(\sqrt{x})$	$\frac{1}{2}$	$\frac{1}{8}$